Evaluating and Training Large Language Models: What Can (and Cannot) Item Parameters Tell Us?

Research Group Seminar · October 26

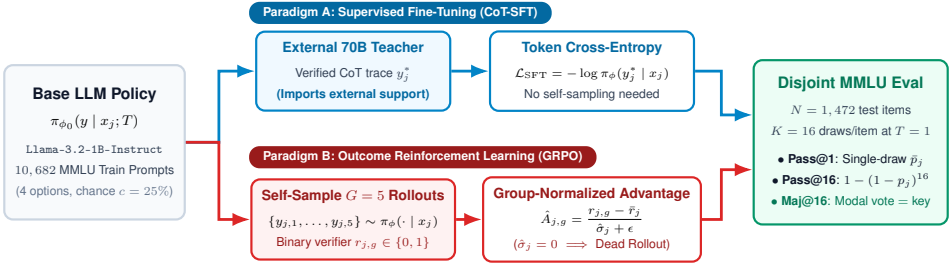
Why Naive 1D Leaderboard 2PL Fails

- **Standard ML Shortcut**: Fit a 1D binary 2PL (a_j, b_j) across 102 leaderboard LLMs and plug it into adaptive testing (`CAT`) or RL critics.
- **Empirical Result**: Hits a $\mathbf{2.4}\%$ bias floor in `CAT` and yields $\mathbf{0}\%$ accuracy gain in RL training.
- **Goal Today**: Diagnose *where and why* it fails.

Pass-Rate Analysis Under an IRT Perspective

- **1. Multidimensionality**: Leaderboard a_j measures cross-model memorization ($5\times$ skew), not reasoning updates.
- **2. Scale vs. Ability**: RL sharpens temperature ($\tau \leftrightarrow 1/T$) rather than shifting ability $\Delta\theta$ ($\mathrm{Maj@16}$ flat).
- **3. Attractive Distractors**: Binary $\{0, 1\}$ scoring is blind to wrong modal distractors below $c = 25\%$ chance.

Background: LLM Post-Training & Evaluation

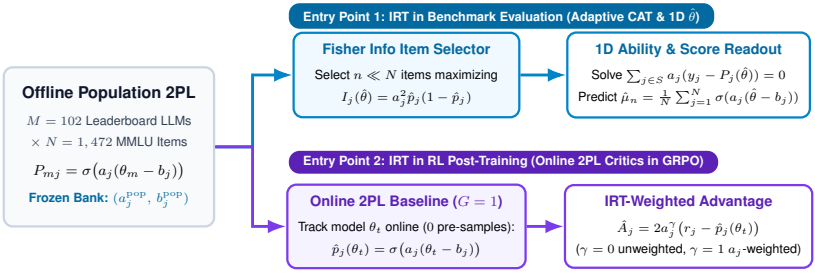


CoT-SFT (Supervised)
Trains on verified 70B teacher traces y_j^*.
→ **Imports external knowledge** even when Base pass rate $p_j = 0$.

GRPO (Outcome RL)
Samples $G = 5$ self-rollouts; rewards binary correctness $r_{j,g} \in \{0, 1\}$.
→ **Zero gradient** if all G rollouts fail or succeed ($\hat{\sigma}_j = 0$).

Test Metrics ($K = 16$ draws)
- $\mathbf{Pass@1}$: Mean single-draw accuracy.
- $\mathbf{Pass@16}$: At least 1 correct in 16 draws.
- $\mathbf{Maj@16}$: Majority-vote accuracy.

Background: How IRT Enters LLM Pipelines



Entry 1: Adaptive Evaluation (CAT)
Calibrate 2PL (a_j, b_j) on $M = 102$ leaderboard LLMs (Maia Polo et al., 2024; Kipnis et al., 2025).
$\rightarrow$ Select high-information items $I_j(\hat{\theta}) = a_j^2 \hat{p}_j(1 - \hat{p}_j)$ to estimate score from $n \ll N$ items.

Entry 2: Online 2PL Critics in RL
Replace $G = 5$ group rollouts with a single rollout ($G = 1$) using IRT expectation $\hat{p}_j(\theta_t) = \sigma(a_j(\theta_t - b_j))$:

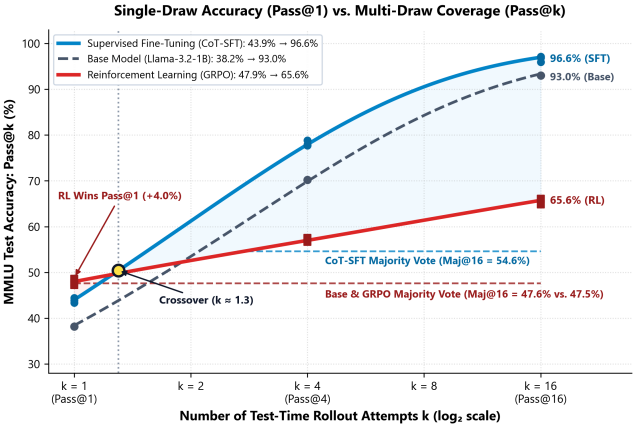
$$\hat{A}_j = 2\, a_j^{\gamma}\big(r_j - \hat{p}_j(\theta_t)\big), \qquad \gamma \in \{0, 1\}$$

Terminology Bridge: Rollout Diagnostics Under an IRT Lens

Concept	LLM Rollout Diagnostic	Psychometric / IRT Perspective	Notation
1. Test-Taker	Model checkpoint π_ϕ ($K = 16$ draws)	Respondent with latent state θ	$\pi_\phi \leftrightarrow \theta$
2. Item Baseline	Base initial pass rate ($p_{\text{base},j}$)	Model-specific item log-odds (μ_{0j})	$\mu_{0j} = \text{logit}(p_{\text{base},j})$
3. Sharpening	Inverse temperature ($1/T$) or RL	Within-respondent dispersion (τ)	$\tau \leftrightarrow 1/T$
4. Chance Pivot	4-choice guessing rate (25%)	Guessing floor (c) vs. distractors	$c = 0.25$
5. Active Variance	`GRPO` rollout variance ($\hat{\sigma}_j^2 > 0$)	Bernoulli / 1PL sampling info	$I_j = p_j(1 - p_j)$
6. Prompt Bias	Option-order gap (`A/B/C/D` swap)	Differential Item Functioning (DIF)	$D = \lvert p_{\text{ord0}} - p_{\text{ord1}} \rvert$

Diagnostic Strategy: When naive 1D population 2PL (a_j, b_j) fails in training, **item-level rollout diagnostics** ($p_{\text{base},j}$, Maj@16, option swaps) let us pinpoint *which psychometric assumptions broke* and why.

The Empirical Puzzle: Pass@1 vs. Pass@k



1. Single-Draw $\mathrm{Pass@1}$ **(**$k = 1$**)**

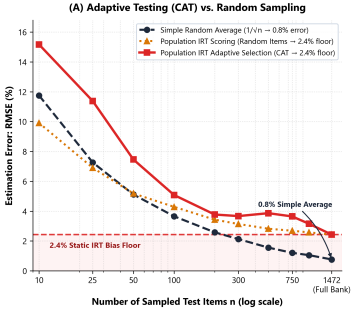
- `Base`: $\mathbf{38.2}\%$
- `CoT-SFT`: $\mathbf{43.9}\%$ $(+5.7\%)$
- `GRPO (RL)`: $\mathbf{47.9}\%$ $(+9.7\%$, wins at $k = 1)$

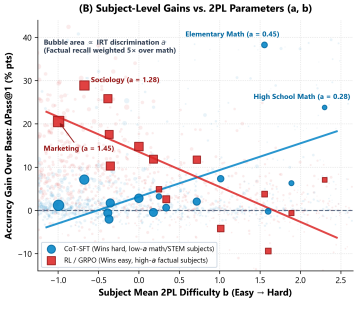
2. Multi-Draw $\mathrm{Pass@16}$ **&** $\mathrm{Maj@16}$

- $\mathrm{Pass@16}$ **collapses under RL**: `GRPO` drops to $\mathbf{65.6}\%$ vs. `Base` $\mathbf{93.0}\%$ and `CoT-SFT` $\mathbf{96.6}\%$.
- **Majority Vote (**$\mathrm{Maj@16}$**) is unchanged by RL**: `GRPO` $(\mathbf{47.5}\%) =$ `Base` $(\mathbf{47.6}\%)$, whereas `CoT-SFT` reaches $\mathbf{54.6}\%$.

$\rightarrow$ **Intuition**: `GRPO` learns **0% net new modal answers**; it only concentrates probability onto existing Base modes.

IRT in Evaluation: Adaptive Testing & Domain Bias





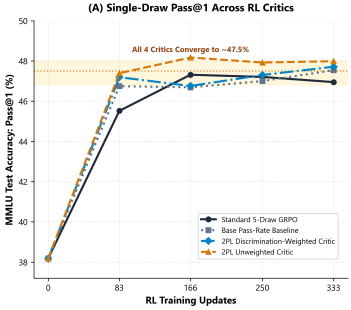
(A) Adaptive CAT Hits a 2.4% Error Floor

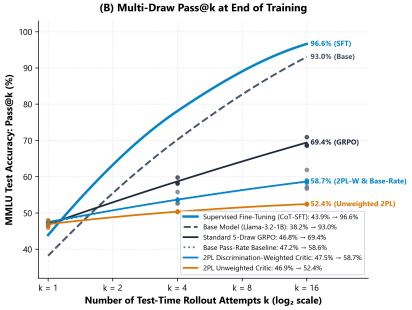
- Simple random sampling reaches $\mathbf{0.8}\%$ RMSE at full bank.
- 1D IRT scoring & CAT plateau at $\mathbf{2.4}\%$ error because post-trained models violate 1D invariance.

(B) $5\times$ Discrimination Skew Across Subjects

- Factual recall has $5\times$ higher a_j (*Marketing* $\mathbf{1.45}$) than math (*HS Math* $\mathbf{0.28}$). Since $I_j \propto a_j^2$, CAT over-samples memory $25\times$.
- GRPO gains on easy/high-a items; CoT-SFT gains on hard/low-a math.

IRT in RL Training: Online 2PL Critics





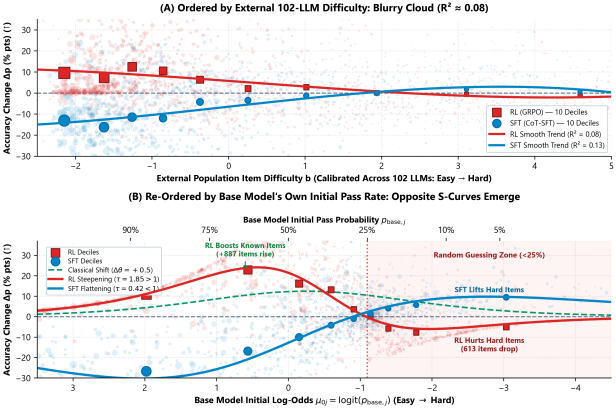
(A) All 4 Critics Saturate at $\sim 47.5\%$ Pass@1

- Standard `GRPO` ($G = 5$) and single-rollout ($G = 1$) 2PL critics all converge to $\mathbf{46.8\%}$–$\mathbf{47.5\%}$.
- Online 2PL baseline cuts sampling cost $5\times$, but a_j-weighting adds **zero accuracy gain**.

(B) High-k Collapse & Item-Subset Divergence

- At $k = 16$, all RL critics collapse ($\mathbf{52.4\%}$–$\mathbf{69.4\%}$) far below `Base` ($\mathbf{93.0\%}$).
- Despite tied aggregate Pass@1, **5 of 6 critic pairs** learn statistically distinct item subsets ($p < 0.05$).

Pass-Rate Analysis: Why Naive 1D 2PL Fails in Training



1. Leaderboard 1D Scale $\neq$ Learning Direction (Panel A)

Ordering item gains Δp_j by 102-LLM difficulty b_j gives a flat cloud ($\mathbf{R^2 = 0.08}$).

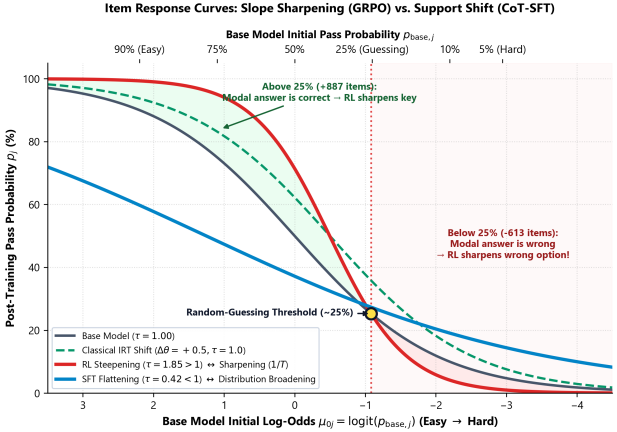
$\rightarrow$ Policy updates are orthogonal to cross-model a_j ($\mathbf{r = 0.072}$): **measuring standing $\neq$ receiving change**.

2. Pass-Rate Diagnostic Frame (Panel B)

Ordering by the Base model's initial log-odds $\mu_{0j} = \text{logit}(p_{\text{base},j})$ reveals two opposite S-curves ($\mathbf{R^2 = 0.35}$):

- **Above 25% chance**: `GRPO` boosts $+887$ already-known items.
- **Below 25% chance**: `GRPO` hurts 613 hard items, whereas `CoT-SFT` lifts them.

Pass-Rate Analysis: Dispersion Sharpening vs. Attractive Distractors



1. Separating Dispersion (τ) from Shift (α)

Diagnostic pass-rate curve against initial log-odds μ_{0j}:

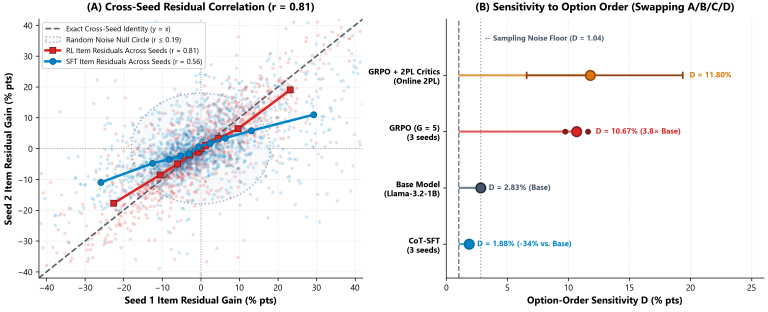
$$p_j = \sigma\big(\alpha + \tau(\mu_{0j} - \mu_c)\big), \quad \mu_c = \text{logit}(0.25)$$

- **Dispersion** $\tau \leftrightarrow 1/T$: Sharpens existing preferences (confounded with $\Delta\theta$ in standard 2PL).
- **Shift** α: Imports new knowledge support.

2. Why Binary $\{0, 1\}$ Scoring Fails Below 25%

- `GRPO` steepens τ ($\mathbf{1.85 > 1}$):
  - When $p_{\text{base},j} > 25\%$, modal answer is correct → sharpening helps.
  - When $p_{\text{base},j} < 25\%$, an **attractive wrong distractor is modal** → binary RL locks in the wrong option ($p_j \to 0$).

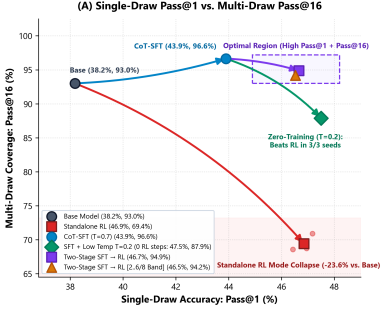
Diagnostics: Cross-Seed Residuals & Option Order

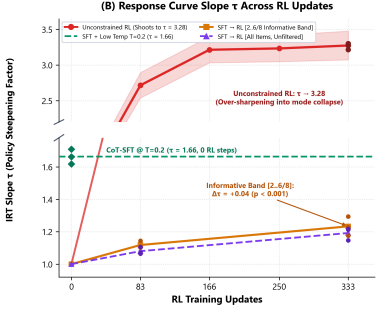


(A) Cross-Seed Residual Correlation ($\mathbf{r = 0.81}$)
After removing the 1D curve, item residuals across random seeds remain highly correlated ($\mathbf{r = 0.81}$ in RL vs. 0.19 noise floor).
$\rightarrow$ Item-level learning is deterministic and multi-dimensional, set by Base initialization.

(B) RL Amplifies Option-Order Bias $4\times$
Swapping option letters (`A/B/C/D`) measures position sensitivity D:
- `Base`: $\mathbf{2.83\%}$ | `CoT-SFT`: $\mathbf{1.88\%}$ (-34% bias)
- `GRPO`: $\mathbf{10.67\%}$ ($4\times$ **position bias** when sharpening low-confidence items).

Implications: Decoding Temperature & Informative Rollouts





1. Temperature Shortcut
Since RL acts like $\tau \leftrightarrow 1/T$, decoding `CoT-SFT` at $\mathbf{T = 0.2}$ (**0 RL steps**) jumps to $\mathbf{47.5}\%$ Pass@1 while keeping $\mathbf{87.9}\%$ Pass@16.

2. Two-Stage SFT → RL
Shifting support first via `CoT-SFT` and then running RL achieves both high Pass@1 ($\mathbf{46.7}\%$) and Pass@16 ($\mathbf{94.9}\%$).

3. Informative Band $[2..6/8]$
Filtering items with variance $p_j(1-p_j) > 0$ cuts dead rollouts $\mathbf{34}\% \rightarrow \mathbf{23}\%$ and steepens τ ($+\mathbf{0.04}, \mathbf{p < 0.001}$).

Summary: Why Naive 1D Binary 2PL Fails & What Diagnostics Show

Failure Mode	Naive 1D 2PL Assumption	Pass-Rate Diagnostic Finding	Psychometric Requirement
1. Memory vs. Reasoning	1D θ captures all capability across subjects.	a_j **skews** $5\times$ **toward memory** (1.45) **vs. math** (0.28) $\rightarrow$ 2.4% `CAT` **floor.**	**Multidimensional IRT (MIRT)** separating memory vs. reasoning.
2. Measure vs. Transition	Cross-model a_j predicts within-model RL updates.	**All 4 critics tie at** $\sim 47.5\%$ $\mathrm{Pass@1}$**; updates orthogonal to** a_j ($\mathbf{r = 0.072}$)**.**	**Decouple Measurement** a_j **from Transition** ψ_j (`DUET`).
3. Ability vs. Temperature	Rising $\mathrm{Pass@1}$ ($38.2\% \rightarrow 47.9\%$) means higher θ.	**RL learns 0% new modes (**$\mathrm{Maj@16} = 47.5\%$**);** `SFT @ T=0.2` **hits** 47.5% **(0 RL).**	**Separate Location (**θ_t**) from Dispersion (**$\tau_t \leftrightarrow 1/T$**).**
4. Binary $\{0, 1\}$ **Scoring**	Wrong answers below 25% are uniform noise.	**Wrong distractors are modal below** $25\% \rightarrow$ **RL hurts 613 items &** $4\times$ **order bias.**	**Nominal Response Model (NRM)** over all 4 options (A, B, C, D).
5. Residual Structure	Residuals are i.i.d. noise after 1D conditioning.	**Cross-seed residuals correlate at** $\mathbf{r = 0.81}$**;** $[2..6/8]$ **band cuts dead rollouts** $34\% \rightarrow 23\%$**.**	**Longitudinal State-Space Calibration** along checkpoints.

Next Steps: Theoretical Base & Proposing DUET

1. Theoretical Base

Build a psychometric model for LLMs that incorporates all 3 missing components:

- **Multidimensional State** $\boldsymbol{\theta}_t \in \mathbb{R}^D$ (memory vs. reasoning).
- **Dispersion Separation**: Decouple location $\boldsymbol{\theta}_t$ from policy temperature $\tau_t \leftrightarrow 1/T$.
- **Nominal Distractor IRT**: Model (p_A, p_B, p_C, p_D) to avoid sharpening wrong distractors below 25%.

2. Proposing DUET

Decoupled Estimation & Transition state-space framework:

- **Observation (a_j)**: Filters static belief $z_t \sim \mathcal{N}(\hat{z}_t, \Sigma_t)$.
- **Transition (ψ_j, b_j)**: Untied practice vector ψ_j + separate receiving subspace ($a_j^\top z_t + b_j^\top w_t$).

$\rightarrow$ Prevents cross-sectional scale $\bar{a}$ from collapsing within-model learning updates.

3. Prior PLMS Lessons [Placeholder]

Design guards from earlier PLMS negative results:

- **Martingale Filter (M1b)**: Zero drift on no-change trajectories.
- **Outcome-Blind Step (M1a)**: Exclude realized y_t from transition step size u_t.
- [Placeholder: Specific PLMS Failure Cases & Null Controls]: Routing deconfounding & $\mathcal{O}(\Delta^2)$ SNR bounds.

Thank You!

Backup B1: First-Order Expansion & Full Results

First-Order Logit Expansion

For policy log-odds $z_j(\phi) = \operatorname{logit} p_j(\phi)$ under parameter update $\Delta\phi = \alpha_\parallel u_{\text{pop}} + u_\perp$:

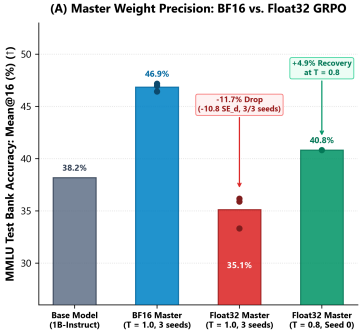
$$\Delta p_j \approx p_{0j}(1 - p_{0j})\left[\alpha_\parallel a_j^{\text{pop}} + J_j u_\perp\right]$$

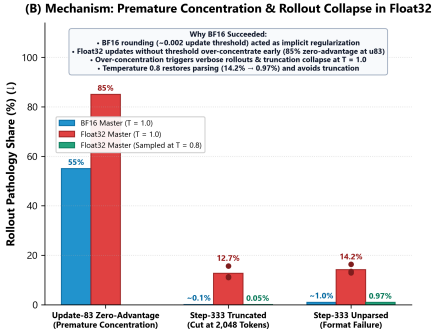
- $\operatorname{corr}(J_j u_\perp, a_j^{\text{pop}}) = \mathbf{0.072}$.
- Within-model variance $p_{0j}(1 - p_{0j})$ drives Δp_j ($R^2 = 0.354$), while a_j^{pop} has near-zero alignment ($r = +0.025$).

Disjoint MMLU Test Bank ($N = 1,472$)

Model / Arm	Draws	Pass@1	Pass@16	Maj@16	Order D
Base (Llama-3.2-1B)	—	38.2%	93.0%	47.6%	2.83%
CoT-SFT (70B Teacher)	0	43.9%	**96.6%**	**54.6%**	**1.88%**
GRPO Standard ($G = 5$)	5	46.9%	69.4%	47.5%	10.67%
Base Pass-Rate Baseline	1	47.2%	58.6%	47.4%	11.42%
2PL Unweighted ($\gamma = 0$)	1	46.9%	52.4%	47.1%	12.15%
2PL a_j-Weighted ($\gamma = 1$)	1	**47.5%**	58.7%	47.5%	11.80%
CoT-SFT @ T=0.2 (0 RL)	0	**47.5%**	87.9%	54.6%	1.95%
Two-Stage SFT → RL	1	46.7%	94.9%	54.4%	3.12%
Two-Stage [2..6/8 Band]	1	46.5%	94.2%	54.5%	2.98%

Backup B2: Numerical Precision Audit (BF16 vs. Float32)





(A) Float32 Master Weights Drop Accuracy by -11.7%

Removing BF16 rounding (~ 0.002 threshold) lowers $\mathrm{Pass@1}$ from $\mathbf{46.9}\% \rightarrow \mathbf{35.1}\%$ (partially recovered at $T = 0.8$).

(B) Premature Concentration in Float32

Exact Float32 updates over-concentrate early ($\mathbf{85}\%$ zero-variance rollouts vs. 55% in BF16), causing truncation (12.7%) and parse errors (14.2%).